

Physics-Exact Credit Assignment for Distributed Flutter Suppression

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Abstract

Active flutter suppression is normally posed as a centralised control problem: one regulator observes the whole wing and commands every control surface. Modern aircraft, however, carry many spanwise-distributed trailing-edge surfaces, each with its own actuator and local sensing, which makes a distributed formulation natural and a centralised one increasingly awkward. We pose flutter suppression as a cooperative multi-agent reinforcement learning problem, with one agent per control surface, local accelerometer sensing, and nearest-neighbour communication. The central difficulty is credit assignment: every surface acts on the same coupled bending–torsion mode, so a shared reward is nearly uninformative about who did what.

Our main observation is that this plant hands us the decomposition that multi-agent reinforcement learning usually has to learn. Differentiating the aeroelastic energy along trajectories isolates a control-power term that is exactly additive across agents, from which the Wolpert–Tumer difference reward follows in closed form: no counterfactual rollout, no learned mixing network, and no centralised critic. We verify the identity numerically against brute-force counterfactuals to 10^{-10} , including the absence of cross-agent terms. We further show that the credit signal must be normalised by the instantaneous energy—the raw form rewards *sustaining* the oscillation, because an agent is paid for having energy available to extract—and that dividing by energy preserves exact additivity while making the credit signal identical to the evaluated objective.

We evaluate on a modally reduced Goland wing with unsteady strip aerodynamics, validated against the published benchmark (first bending 7.664 Hz versus 7.66 Hz; flutter at 137.35 m s^{-1} versus 137 m s^{-1}), and against a four-rung classical baseline ladder that includes both centralised and fully decentralised LQG, plus a fault-aware oracle used only to establish which test conditions are feasible at all. *[ResultsSummarySentence: to be written]*

1 Introduction

Flutter is a self-excited aeroelastic instability in which structural bending and torsion couple through unsteady aerodynamic loads and extract energy from the freestream. Beyond a critical speed the coupled mode becomes unstable and the response grows without bound. It is a hard structural limit on the flight envelope, and because it is a dynamic instability rather than a static one it can in principle be pushed back by active control.

Active flutter suppression has an extensive classical literature built on linear-quadratic and H_∞ synthesis, and a growing reinforcement-learning literature. The learning work is, to our knowledge, uniformly single-agent: a single policy commands a single effector, whether that is a synthetic jet, trailing-edge circulation control [2], camber morphing [14], or the parameters of a conventional controller [1]. Meanwhile the hardware trend runs the other way. A modern flexible wing carries several independently actuated trailing-edge surfaces, each with local sensing and local computation, connected by a bus rather than to a central controller.

This paper takes that hardware structure seriously and asks the question it implies: not *which* multi-agent algorithm to apply, but what the physics of a coupled continuum instability tells us about credit assignment and communication that a generic multi-agent algorithm would otherwise have to discover.

Three properties of the plant make generic approaches a poor fit.

1. **The instability is modal, not spatial.** A specific coupled bending–torsion eigenvalue crosses into the right half plane. Each surface acts locally in space, but its effect on that mode is set by the modal residue at its station. Two agents can read identical local sensors and have opposite influence.
2. **A shared scalar reward is nearly uninformative.** All surfaces act on the same mode and the damping they produce is a sum. Learned decompositions [18, 15, 9] must infer the split from noisy returns.
3. **Authority is signed and condition-dependent.** For a section whose elastic axis lies aft of the aerodynamic centre, a trailing-edge-down deflection produces lift up *and* a nose-down twist; which term dominates depends on dynamic pressure, so the sign of an agent’s useful action can invert across the envelope.

Contributions.

- A closed form for the multi-agent difference reward in an energy-conserving structural plant with additive control authority (Proposition 1), verified numerically against brute-force counterfactuals, together with the observation that the credit signal must be the *logarithmic* decay rate rather than the raw energy rate (Section 4.2).
- MoCCA, a distributed policy architecture combining that credit signal with a fixed-bandwidth modal-phasor consensus channel and a phase-locked action parameterisation.
- A validated aeroelastic testbed and a four-rung classical baseline ladder that separates the effect of the control method from the effect of the information structure available to it, plus a feasibility oracle that identifies test conditions no controller can hold.

2 Related work

Learning for flutter control. Recent work applies deep reinforcement learning to stall flutter via camber morphing [14], to flutter suppression by trailing-edge circulation control validated in a wind tunnel [2], and to tuning aeroservoelastic controller parameters on a flexible flying wing [1]. All are single-agent, with one effector and a global observation. We are not aware of prior work that treats the individual control surfaces as separate learning agents.

Multi-agent credit assignment. The standard tools are difference rewards [20] and value factorisation [18, 15], with COMA [9] implementing a counterfactual advantage baseline; recent surveys review the space [5, 4] and work continues on sharper estimators [3]. These methods are general precisely because they assume nothing about the environment. Our point is narrower and complementary: in this class of physical system the decomposition is available analytically, so it need not be estimated at all.

Learned communication. CommNet [17] and TarMAC [7] learn message contents end to end. We instead fix the message *space* to a modal phasor, which makes the channel interpretable, bounds its width independently of the number of agents, and lets its degradation be characterised.

Table 1: Validation against the published Goland wing.

Quantity	This model	Published
First bending frequency	7.664 Hz	7.66 Hz
First torsion frequency	15.232 Hz	15.24 Hz
Flutter speed	137.35 m s^{-1}	$\approx 137 \text{ m s}^{-1}$
Flutter frequency	69.3 rad s^{-1}	$\approx 70.7 \text{ rad s}^{-1}$
Static divergence speed	$356.8 \text{ m s}^{-1} (2.60 U_f)$	—

3 Plant and problem formulation

3.1 Aeroelastic model

We use a modally reduced uniform cantilever wing. The structure is a Galerkin projection onto exact clamped-free bending and fixed-free torsion modes, with all generalised mass and stiffness integrals evaluated by Gauss–Legendre quadrature. Aerodynamics are strip theory carrying the full Theodorsen non-circulatory terms [19], with the circulatory lag realised in the time domain by two Wagner lag states per strip using the two-exponential approximation [12]. Control surfaces contribute quasi-steady thin-airfoil flap derivatives from the Glauert series, integrated over each surface’s spanwise band. The resulting state is

$$z = \begin{bmatrix} q & \dot{q} & x_w \end{bmatrix}^\top, \quad M\ddot{q} + C\dot{q} + Kq = Lx_w + B\delta + b_g w_g, \quad (1)$$

with q the modal amplitudes, x_w the wake lag states, δ the surface deflections and w_g a vertical gust. Plunge is positive down and twist positive nose-up, following the classical convention [10, 6] that keeps the unsteady operators in textbook form.

3.2 Validation

Table 1 compares the model against the published Goland cantilever wing [11]. Agreement is to better than 0.5 % on the modal frequencies and on the flutter speed. Every reported result is pinned to this benchmark by an automated test rather than to a value the implementation once produced.

3.3 Distributed control problem

Three trailing-edge surfaces occupy spanwise bands at $[0.05, 0.40]$, $[0.40, 0.72]$ and $[0.72, 0.98]$ of the semispan, hinged at 75 % chord, with $\pm 15^\circ$ travel, 200° s^{-1} rate limit and a 20 ms first-order servo lag. Each agent observes only a forward/aft accelerometer pair at its own station, its local plunge rate, twist and twist rate, its own deflection, and air data. No agent observes the modal state: the coupled flutter mode is a global quantity that no single station can measure, and that partial observability is the problem rather than an artefact of our setup. Control runs at 200 Hz.

4 Method

4.1 The energy budget and its exact decomposition

Take the aeroelastic energy $E = \frac{1}{2}\dot{q}^\top M\dot{q} + \frac{1}{2}q^\top Kq$ and differentiate along trajectories of (1):

$$\dot{E} = \underbrace{-\dot{q}^\top C\dot{q}}_{\text{dissipation}} + \underbrace{\dot{q}^\top Lx_w}_{\text{wake exchange}} + \sum_k \underbrace{(\dot{q}^\top b_k) \delta_k}_{P_k}, \quad (2)$$

where b_k is the k th column of B . The final term is exactly additive across agents, and P_k is agent k 's instantaneous control power; $P_k < 0$ means surface k removes energy from the structure.

Proposition 1 (Closed-form difference reward). *For a global objective whose only agent-dependent term is the control power in (2), the difference reward $D_k = G(z) - G(z_{-k})$ admits the closed form $D_k = -P_k$. No counterfactual rollout, learned mixing network, or centralised critic is required to evaluate it.*

Proof. Removing agent k sets $\delta_k = 0$. Every other term of (2) is independent of δ_k , and the control-power term is a sum over agents, so the difference collapses to the single term $(\dot{q}^\top b_k)\delta_k$. $\square$

Projecting onto the in-vacuo modes ($q = V\eta$) keeps the separation in both agent and mode: agent k 's contribution to mode r is $\dot{\eta}_r(v_r^\top b_k)\delta_k$. The factor $v_r^\top b_k$ is the modal residue, the quantity that decides whether a surface can damp or will excite that mode, and whose sign inverts under control reversal.

4.2 The credit signal must be the logarithmic decay rate

Used directly, $-P_k$ fails, and instructively. It is maximised by having energy available to extract, so an agent is rewarded for *sustaining* the oscillation. Policies trained on it converge to holding the wing at a bounded but undamped amplitude: we measured $+0.265\text{ s}^{-1}$ (energy slowly growing) against -0.06 s^{-1} for a hand-tuned collocated damper. This is reward hacking, not merely myopia.

The remedy follows from the same identity. Dividing (2) by E ,

$$\frac{d}{dt} \log E = \frac{-\dot{q}^\top C \dot{q} + \dot{q}^\top L x_w + \sum_k P_k}{E}, \quad r_k = -\frac{P_k}{E + \varepsilon}. \quad (3)$$

Because E is a shared scalar the decomposition survives division exactly, so Proposition 1 carries over to the log-decay rate. Three properties follow. The signal is scale invariant, so there is no incentive to sustain the oscillation. It coincides with the evaluated objective, since the reported metric is the exponential energy decay rate, which is $\frac{1}{2}d(\log E)/dt$. And it remains exactly additive. Switching to (3) inverted the sign of the outcome, from $+0.265$ to -0.291 s^{-1} , with training divergence falling from 0.8 % to 0.0 %.

Remark 1. *The exact signal is still myopic: cross-mode coupling through C and through the wake states means an agent may have to inject energy into one mode so that another can extract more from the coupled pair. We therefore add potential-based shaping [13] on $\log E$, which shares the credit signal's units and, being potential-based, cannot change which policy is optimal. Blending instead with an energy level, as we first did, is a scale error: the two terms differed by roughly $50\times$ and the shared term swamped the credit.*

4.3 Spectral consensus and phase-locked actions

Agents do not exchange observations. Each runs a recurrent encoder over its own accelerometer history producing a phasor estimate for R retained critical modes, and the agents then run T rounds of averaging consensus over the spanwise line graph—nearest neighbours only, which is what a distributed avionics bus physically is. The message is $2R$ numbers per agent per step, independent of the number of surfaces.

Actions are emitted as a gain and a phase relative to the consensus phasor. For a mode with phasor (a_r, b_r) , acting with gain A_k^r at offset ψ_k^r means $A_k^r(a_r \cos \psi_k^r - b_r \sin \psi_k^r)$, a scaled rotation of the phasor. This bakes the resonant structure of flutter suppression into the action space. The resonant term is *added* to a general action channel rather than replacing it, so the

head is a strict generalisation of an unconstrained policy; our first version replaced the general action, making the parameterisation a restriction, and the resulting policy scored worse than its own ablation.

4.4 Training

We use shared-parameter multi-agent PPO [16, 21] with per-agent advantages, since with the physics credit each agent receives a different reward. Updates run on *sequences*: minibatches are segments of consecutive steps unrolled with gradient from a detached stored hidden state. Sampling shuffled individual timesteps, as we did initially, trains the recurrence as a one-step map and gives the phasor encoder no temporal gradient at all, which is precisely what it needs.

5 Experimental design

5.1 A baseline ladder that separates method from information

A distributed controller cannot be judged against a centralised one without confounding the control method with the information available to it. Decentralisation costs performance; that is a known result in distributed control, not a defect. We therefore report four rungs: collocated local rate feedback (no model, no communication); gain-scheduled LQR on the *full* 56-state plant including the wake states (not implementable, a ceiling); centralised LQG driven by a Kalman observer on the same six accelerometer channels the agents see; and a fully decentralised LQG in which each surface runs its own observer on its own two accelerometers and commands only itself. All are synthesised in discrete time at the control rate. The last rung is the honest target.

5.2 Feasibility

We additionally run a fault-aware oracle: a gain bank re-synthesised for each failure hypothesis, given the full state and told which surface has failed. Its role is not competition but feasibility. Of the nine evaluation cells, 6 are stabilisable by this oracle; the remainder are stabilisable by no controller with these actuators, and are reported as infeasible rather than as failures of any method.

5.3 Metrics

PPO return cannot rank runs whose reward functions differ, so all controllers are scored on the same physical quantities over identical episodes—same airspeeds, initial deformation, turbulence and injected failure: the fraction of episodes that leave the linear regime, the exponential energy decay rate, peak tip plunge, and RMS deflection.

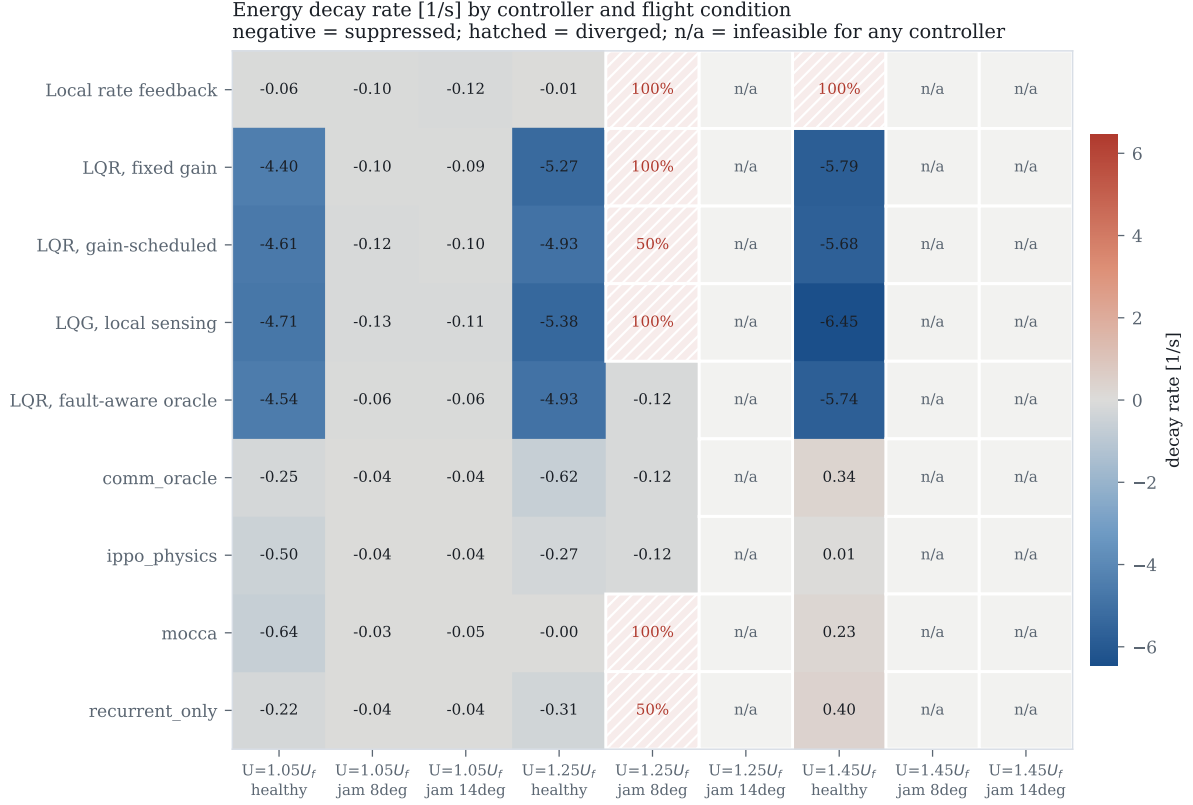


Figure 1: Energy decay rate by controller and flight condition. Negative is suppressed; hatched cells diverged; *n/a* marks conditions infeasible for any controller, established by the fault-aware oracle.

6 Results

Condition	local_fb	lqr_fixed	lqr_sched	lqg_local	lqr_oracle	comm_oracle	ippo_
U=1.05U _f healthy	-0.06	-4.40	-4.61	-4.71	-4.54	-0.25	
U=1.05U _f jam#2@8deg	-0.10	-0.10	-0.12	-0.13	-0.06	-0.04	
U=1.05U _f jam#2@14deg	-0.12	-0.09	-0.10	-0.11	-0.06	-0.04	
U=1.25U _f healthy	-0.01	-5.27	-4.93	-5.38	-4.93	-0.62	
U=1.25U _f jam#2@8deg	D100%	D100%	D50%	D100%	-0.12	-0.12	
U=1.25U _f jam#2@14deg	—	—	—	—	—	—	
U=1.45U _f healthy	D100%	-5.79	-5.68	-6.45	-5.74	0.34	
U=1.45U _f jam#2@8deg	—	—	—	—	—	—	
U=1.45U _f jam#2@14deg	—	—	—	—	—	—	
Stabilised	4/6	5/6	5/6	5/6	6/6	6/6	

[ResultsNarrative: to be written]

7 What coordination looks like

“Emergent coordination” is easy to assert and hard to pin down, so we define it as a measurable quantity: the phase at which each surface acts relative to the critical modal motion, as a function of spanwise station. The measurement is the cross-spectrum of each surface deflection against the first modal velocity, evaluated at the frequency where the modal response peaks, and it is applied identically to learned policies and to classical controllers.

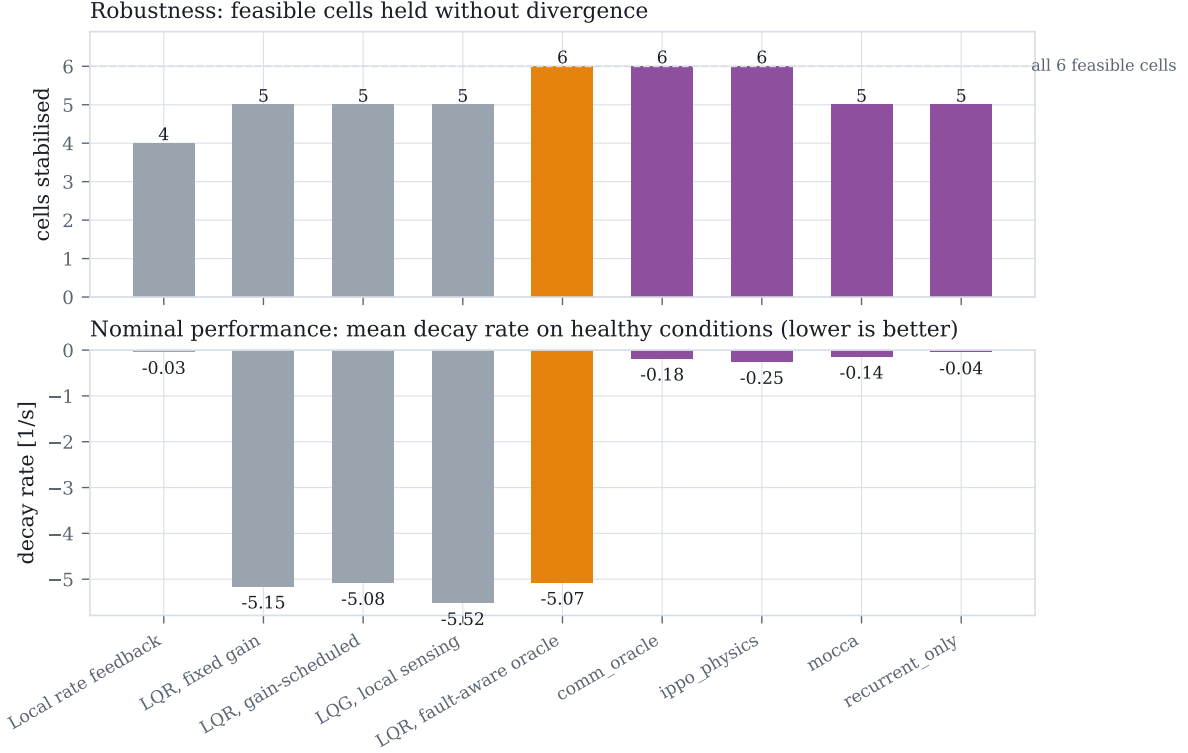


Figure 2: Robustness (top) and nominal performance (bottom). Robustness counts feasible conditions held without divergence.

Table 2: Spanwise actuation phase of the classical controllers. The gradient is the least-squares slope of phase against span.

Controller	Dominant frequency	Phase gradient
Local rate feedback	5.50 Hz	-15° per semispan
Decentralised LQG	7.50 Hz	27° per semispan
Centralised LQG	11.00 Hz	90° per semispan

Applied to the classical ladder it already separates the rungs sharply, which is what makes it a useful instrument rather than a decoration.

Two things stand out. The more capable the controller, the stronger its spanwise phase progression: collocated feedback is essentially flat, meaning every surface acts at the same phase and no spanwise pattern exists, while centralised LQG develops roughly 90° of phase across the semispan. And only the centralised controller operates at the flutter frequency itself (11.00 Hz, against the 11.03 Hz predicted by the eigenvalue analysis); local feedback responds at 5.50 Hz and never engages the unstable mode at all, which is the mechanism behind its poor performance rather than a separate observation about it.

This gives the learned policies something falsifiable to be held against: a distributed policy that recovers a comparable phase progression has discovered, from local measurements alone, the coordination pattern that optimal centralised synthesis arrives at with the full model and the full state.

[PhaseNarrative: to be written]

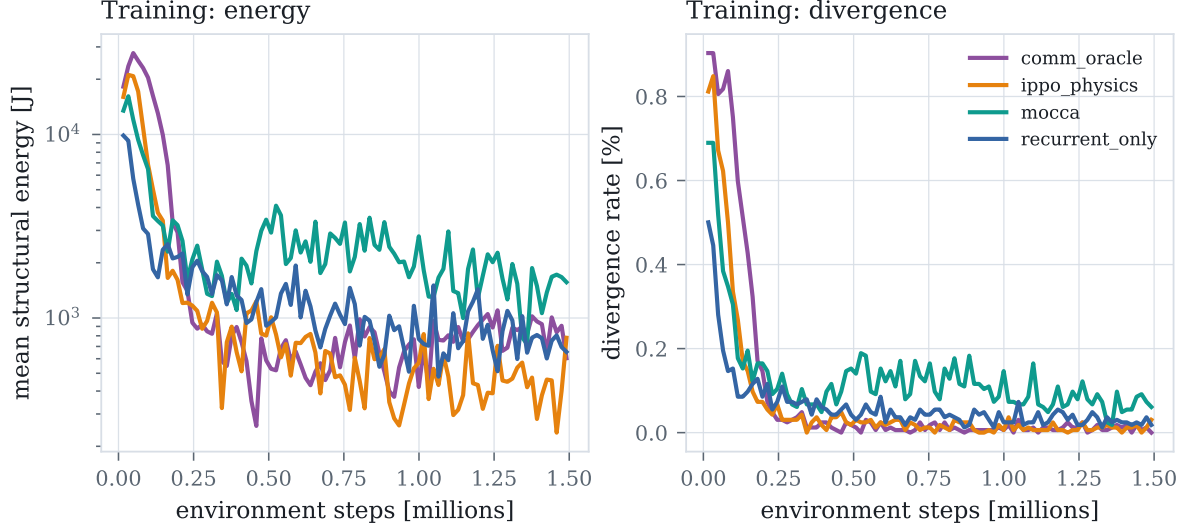


Figure 3: Training curves. The task distribution is ramped over the first 40 % of training, which is visible as a rise in energy as harder flight conditions and failures enter the sampling.

8 Ablations

[ResultsAblations: to be written]

9 Limitations

We state these plainly.

- The credit signal requires the modal control influence $v_r^\top b_k$. This is free in simulation but on hardware needs a modal model; robustness to error in B is an open item.
- Quasi-steady flap aerodynamics neglect the flap’s own wake shedding, which over-estimates control authority at high frequency. Re-validating a trained policy on a higher-fidelity unsteady vortex-lattice model such as SHARPy [8] is the natural next step and has not been done here.
- Proposition 1 concerns the difference reward, not optimality of the resulting policy; the sufficiency of the phasor channel is a truncation argument, not an equivalence, and we do not claim otherwise.
- Results are on a three-surface wing with two seeds per variant. The scaling claim implied by the fixed-bandwidth channel is untested beyond three agents.
- “Emergent” coordination is used here only in the operational sense of a spanwise phase pattern that was not specified in advance.

10 Conclusion

Posing flutter suppression as a distributed multi-agent problem exposes a credit assignment structure that the physics resolves in closed form: the difference reward is the agent’s own control power, normalised by the instantaneous energy so that it coincides with the objective and cannot be gamed by sustaining the oscillation. *[ConclusionSentence: to be written]*

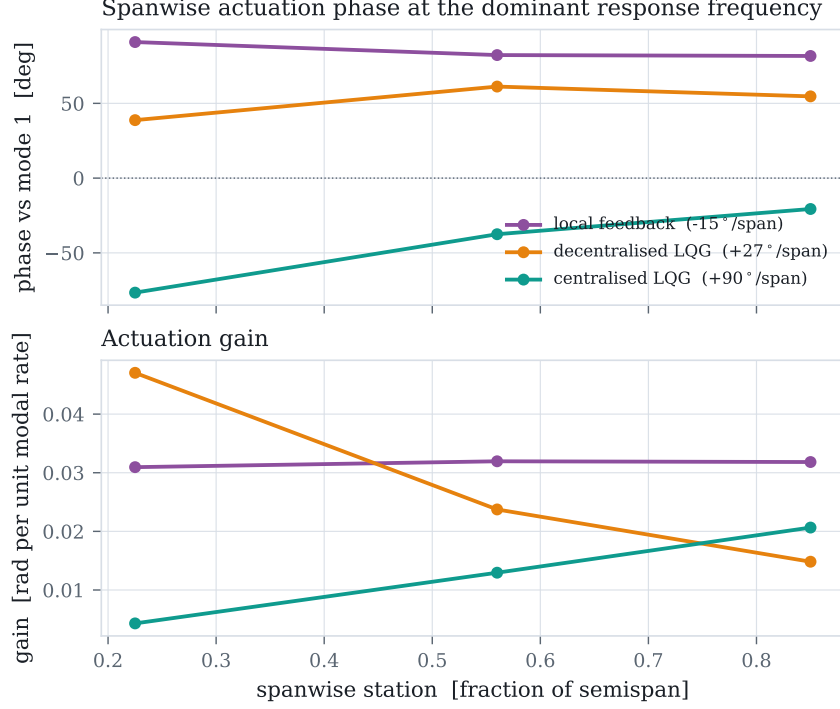


Figure 4: Spanwise actuation phase and gain at the dominant response frequency. A sloped phase profile is a travelling-wave actuation pattern matched to the mode shape; a flat profile is purely local reaction.

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